KEYNOTE TOOLS

User Guide & Technical Reference

Bear Engineering | AppleScript Automation Suite | v2.0

4 Programs. One Tool. Plus Kurt's Version.

Automatically organizes photos, places arrows, creates Item Image labels, handles overflow onto continuation slides, and condenses slides in Keynote inspection reports. Kurt's Version preserves manually arranged photo order.

Overview

Keynote Tools is an AppleScript program that automates the formatting of Keynote-based foundation and drainage inspection reports at Bear Engineering. It eliminates the manual, repetitive work of positioning photos, creating Item Image labels, cleaning up text boxes, sizing body text, placing directional arrows, handling photo overflow onto continuation slides, and consolidating sparse slides.

The tool contains four programs accessible from a single menu when the script is run. It is designed to be used by inspectors with no programming knowledge: launch it, pick a program, confirm the file, and let it run.

A separate variant called Kurt's Version is also available. It has the same three core programs but modifies Program 1 to respect the inspector's manual photo arrangement rather than reassigning photos sequentially.

Design Intent

Inspection reports follow a consistent Keynote template: content slides each containing up to nine site photos, a body text box with observed conditions, and a slide title. Photos arrive from the field in arbitrary positions and must be repositioned into a defined grid before the report goes out.

The tool was designed around four distinct workflows:

- **Program 1:** A full pass over a fresh report that needs everything done at once.
- **Program 2:** A quick re-do of a single slide after edits, with arrows.
- **Program 3:** A re-straightening of a single slide where the inspector has manually arranged photos, preserving order without placing arrows.
- **Program 4:** A cleanup pass after content editing has left some slides partially filled.

Programs 1 and 2 place directional arrows on each photo. Program 3 does not. Use Program 3 when you want to fix photo positions after already having placed arrows, so arrows are not duplicated.

Required Files and Folder Structure

The Script

Keynote Tools.applescript is stored on the Desktop. Open it in Script Editor and click Run each time you want to use it.

The Arrows Folder

Programs 1 and 2 place PNG arrow images onto photos. These files must be stored in a subfolder called Arrows inside the same folder as the Keynote report being processed.

Required folder structure:

```
Your Job Folder/
    Foundation & Drainage Inspection Report_123 Main St.key
    Arrows/
        red_arrow_horiz.png
        red_arrow_vert.png
        blue_arrow_horiz.png
```

The Arrows folder must be named exactly "Arrows" (capital A) and must be a direct subfolder of the folder containing the Keynote file. If a report is moved, the Arrows folder must move with it.

File	Description
red_arrow_horiz.png	Default arrow used on most slides. Points horizontally toward the area of concern.
red_arrow_vert.png	Used on concrete cracking and rebar rusting slides. Points downward.
blue_arrow_horiz.png	Used on drainage and exterior grade/slope slides. Points horizontally in blue.

Installation for a New Inspector

1. Copy Keynote Tools.applescript to your Desktop.
2. Open it in Script Editor (search for Script Editor in Spotlight). You should see the full script text.
3. Copy the Arrows folder (containing the three PNG files) into each job folder alongside the Keynote report. Keep a master copy on your computer and duplicate it into each new job folder as needed.
4. The first time you run Program 4, macOS will ask for Accessibility permission. Go to System Settings → Privacy & Security → Accessibility and enable Script Editor. This only needs to be done once.
5. To run the tool, open Keynote Tools.applescript in Script Editor and click the Run button (■). A menu will appear asking which program to run.

The script does not need to be compiled or exported as an app. Running it directly from Script Editor is the intended method.

How to Use the Tool

Each time you run the script, a program selection menu appears. Choose the program you want and click OK.

Open File Detection

If a Keynote file is already open when you run the script, the tool will detect it and ask if you want to use that file. Click “Use This File” to proceed, or “Choose Different File” to browse for a different one.

Page Number Detection

Programs 1 and 4 auto-detect the first and last content slides by finding the slides containing the 200×267px property overview photo. Programs 2 and 3 default to whichever slide is currently visible in Keynote. Hidden slides are excluded from detection. The program runs immediately using the detected values without asking for confirmation.

The Four Programs

Program 1 — Organize All Slides (with arrows)

Run this on a fresh report before sending it out. It processes every slide in the auto-detected range and performs the following passes in sequence:

Pass 1: Photo Layout

Photos are collected from each slide and placed into a standardized grid. Wide photos (434×267px) are always placed last. The grid adapts to the number and type of photos:

Photos	Layout
1–3	Top row (portrait grid, positions 1–3).
4–6	Top two rows.
7–9	All three rows.
Landscape (333×250px)	Two-column landscape grid for all photos on the slide.
Wide (434×267px)	Placed last; skip column 3 so they display correctly across two slots.
More than 9 (portrait) or 6 (landscape)	Overflow photos go onto a continuation slide automatically (see Overflow Handling below).

Pass 2: Item Image Label Creation

After photos are placed, Item Image labels are created for every photo that does not already have one. Labels are numbered sequentially starting at Item Image 1. On overflow continuation slides, numbering continues from where the previous slide left off (e.g. Item Image 10, 11, 12...). Orphaned labels (no photo above them) are deleted.

Pass 3: Text Box Cleanup and Repositioning

Stale text boxes below y=880 are deleted. The main body text box is moved and resized to fill the space below the photos. On rebar rusting slides the body text is set to 535px wide. When a wide photo is present the body text is always placed full-width at the start of the next row down. If the body text would overflow the page, it is moved to a text-only continuation slide instead.

Pass 4: Arrow Placement

An arrow is placed centered on each photo. Arrow type is chosen based on the slide title:

Slide Type	Arrow
Drainage / slope slides	Blue horizontal arrow.
Rebar rusting slides	Red vertical arrow (points downward from center).
All other slides	Red horizontal arrow (default).

Overflow Handling

When a slide has more photos than available grid slots, or when the body text box is too large to fit on the same slide as the photos, the script automatically creates continuation slides:

- The original slide content is copied and pasted onto a new blank slide inserted immediately after.
- Photos already placed on the original slide are deleted from the continuation slide.
- Overflow photos are placed into grid slots on the continuation slide starting at position 1.
- New Item Image labels are created with numbers continuing from the prior slide.
- Arrows are placed on all newly placed photos.
- If there are still more overflow photos, the process repeats until all photos are placed.
- The body text box appears only on the last slide in the group. On all prior slides it is hidden.
- Continuation slide titles append " (cont'd)" automatically.

Slides titled "Foundation plan showing observed conditions" are skipped automatically by Programs 1 and 2.

Program 2 — Fix One Slide (reset photo order, with arrows)

Performs the same photo layout, label creation, text box cleanup, overflow handling, and arrow placement as Program 1, but on a single slide only. Photos are reassigned sequentially to grid positions. The default slide is whichever slide is currently open in Keynote.

This program places new arrows on every photo. If arrows were already placed on the slide, remove them first to avoid duplicates.

Program 3 — Fix One Slide (keep photo order, no arrows)

Performs photo layout and text box cleanup on a single slide without placing arrows. Each photo snaps to its nearest valid grid position rather than being reassigned sequentially, so any intentional manual arrangement is preserved. Item Image labels are created for any photo that does not already have one.

If the body text box is too large to fit on the same slide after repositioning, a text-only continuation slide is created automatically with the body text at the top and the title marked "(cont'd)".

Use Program 3 after manually tweaking photo positions when you want the grid tidied without disturbing your ordering or adding arrows.

Program 4 — Condense Slides

Scans the auto-detected slide range top-to-bottom and merges any slide whose content fits onto the previous slide. For each slide, the script measures the total content height and the bottom edge of the previous slide. If the content fits below with a 10pt overlap gap, it shifts everything down and pastes it onto the previous slide, then deletes the now-empty slide.

Page numbers (x > 700, y > 950) are excluded from the merge and any duplicate page numbers created by the paste are deleted automatically.

Program 4 uses clipboard keystrokes (Cmd+A, Cmd+C, Cmd+V) to copy content between slides. Keynote must remain the frontmost application while it runs. Do not switch to another app until it finishes.

Kurt's Version

Kurt's Version is a separate script (Kurts_Version.applescript) designed for inspectors who prefer to arrange photos manually in Keynote before running the cleanup tool. It contains three programs instead of four.

How It Differs from the Standard Version

The key difference is in Program 1. In the standard version, Program 1 assigns photos to grid slots sequentially in Keynote's internal object order. In Kurt's Version, Program 1 reads each photo's current position on the slide and sorts them by visual location (top-to-bottom, then left-to-right within a row) before snapping each one to its nearest available grid slot.

This means if Kurt has manually placed Photo A in the upper-left and Photo B in the lower-right before running the script, that arrangement is preserved after the cleanup pass rather than being overwritten by Keynote's arbitrary internal order.

Wide photos (434×267px) are still placed last regardless of their manual position, because they require specific slot validation to display correctly.

Programs in Kurt's Version

Program	Name	Description
1	Organize All Slides (with arrows)	Full pass over all slides. Preserves manual photo order by sorting photos by on-slide position before snapping to grid. Places arrows. Handles overflow and creates continuation slides.
2	Fix One Slide (keep photo order, no arrows)	Single-slide cleanup that snaps photos to nearest grid position without reordering or placing arrows. Same as Program 3 in the standard version.
3	Condense Slides	Merges partially filled slides onto the previous slide. Identical to Program 4 in the standard version.

Program 2 from the standard version (Fix One Slide, reset photo order, with arrows) is not included in Kurt's Version, since resetting order is contrary to Kurt's workflow.

When to Use Which Version

- Use the standard Keynote Tools when photos can be dropped anywhere and the script should handle all positioning from scratch.
- Use Kurt's Version when the inspector has already arranged photos in a deliberate order and wants the script to tidy the grid and add arrows without disturbing that arrangement.

Installation

Kurt's Version uses the same Arrows folder and file structure as the standard version. Save Kurts_Version.applescript to the Desktop, open it in Script Editor, and click Run. The program selection menu will display "Kurt's Version" in the prompt so it is clearly distinguishable from the standard script.

Photo Size Reference

The script identifies photo types by their pixel dimensions in Keynote. These sizes must match exactly for the layout logic to work correctly.

Size	Type	How it's used
200 × 267 px	Property overview photo	Marks the first and last content slides for auto-detection of slide range.
333 × 250 px	Landscape site photo	Triggers the landscape grid layout for the whole slide.
434 × 267 px	Wide site photo	Placed last in layout; skips column 3; forces full-width body text box placement.
≥200 px both dimensions	Standard site photos	Placed sequentially in the portrait grid.

Grid Position Reference

Portrait Grid (standard photos)

	Col 1	Col 2	Col 3
Row 1	(50, 57)	(284, 57)	(518, 57)
Row 2	(50, 357)	(284, 357)	(518, 357)
Row 3	(50, 657)	(284, 657)	(518, 657)

Landscape Grid (landscape photos)

	Col 1	Col 2
Row 1	(41, 73)	(394, 73)
Row 2	(41, 373)	(394, 373)
Row 3	(41, 673)	(394, 673)

Item Image Label Positions

Labels are placed at fixed coordinates below each photo slot, independent of photo position.

Grid Slot	Label Position (x, y)
Portrait 1, 2, 3	(50, 330), (284, 330), (518, 330)
Portrait 4, 5, 6	(50, 630), (284, 630), (518, 630)
Portrait 7, 8, 9	(50, 930), (284, 930), (518, 930)
Landscape 1, 2	(50, 330), (400, 330)
Landscape 3, 4	(50, 630), (400, 630)
Landscape 5, 6	(50, 930), (400, 930)

Body Text Box Sizing

The body text box is identified as the text box with more than 100 characters. Its position and width are set based on photo count after layout. When a wide photo is present, the normal side-placement logic is bypassed entirely. If the text box would extend below y=1020, it is moved to a continuation slide.

Photo count	Width (normal)	Width (rebar)	Position
Wide photo present (any count)	700 px	700 px	x=34, y=next row + 20pt
0, 3, or 6	700 px	535 px	x=34, y=next row start
1, 4, or 7 (short text)	465 px	465 px	x=269, y=bottom row start
1, 4, or 7 (long text >2000 chars)	700 px	535 px	x=34, y=next row start
2, 5, or 8 (short text)	232 px	232 px	x=502, y=bottom row start
2, 5, or 8 (long text >650 chars)	700 px	535 px	x=34, y=next row start

Slide Title Detection

The arrow type and body text width are determined by reading the slide title. The script identifies the title as the text box whose top edge (y position) is less than 50 points from the top of the slide canvas. Two specific title strings trigger special behavior:

Title contains…	Effect
“Observed Condition - Drainage Improvements - Exterior grade/slope”	Blue horizontal arrow is used instead of red.
“Observed Condition - Concrete cracking/spalling from steel rebar rusting”	Red vertical arrow is used, and body text box width is reduced to 535px.
“Foundation plan showing observed conditions”	Slide is skipped entirely by Programs 1 and 2 (and Program 1 in Kurt’s Version).

Title matching uses “contains” rather than an exact match, so minor variations in surrounding text will not break detection as long as the key phrase is present.

Known Limitations

- Program 4 (and Program 3 in Kurt's Version) requires Keynote to remain frontmost throughout its run, as it uses keyboard shortcuts to copy and paste content between slides. Switching to another application mid-run will cause it to fail or paste into the wrong place.
- Arrow images are placed but not scaled by the script. If the arrow PNG files are resized externally, placement positions may shift.
- The rebar rusting slide title must contain the exact phrase shown above for the narrower body text width and vertical arrow to activate.
- Hidden slides are excluded from auto-detection of start and end pages but are not excluded from Program 1 and 4 processing once the range is confirmed.
- Programs 1 and 2 place arrows on all on-canvas photos each run. Remove existing arrows before re-running to avoid duplicates.
- In Kurt's Version, if two photos are manually positioned closest to the same grid slot, the one that sorts first (higher on the slide) claims that slot and the other is snapped to its next-nearest available slot.

Distributing to Other Inspectors

The script is shared as a plain .applescript file and run from Script Editor. Exporting as a compiled app or Automator application introduces macOS permission issues on newer systems and is not recommended.

1. Send them Keynote Tools.applescript (and Kurts_Version.applescript if applicable).
2. Send them the Arrows folder with all three PNG files.
3. Tell them to save the script to their Desktop, open it in Script Editor, and click Run each time they want to use it.
4. Have them place a copy of the Arrows folder inside each job folder alongside the Keynote report.

Version History

Version	Changes
Kurt'sVersion	Added as a companion script alongside the standard Keynote Tools. Program 1 modified to sort photos by their current on-slide position (top-to-bottom, left-to-right) and snap each to its nearest available grid slot, preserving manual arrangement. Program 2 (reset photo order) removed. Programs renumbered: 1 = Organize All Slides, 2 = Fix One Slide (keep order, no arrows), 3 = Condense Slides.
v2.0	Programs 1 and 2: Full overflow handling — when more photos exist than available grid slots, continuation slides are created automatically with sequential Item Image labels (continuing from where the prior slide left off) and arrows placed on all new photos. Body text moves to the final slide in the group; if it would overflow the page a text-only continuation slide is created. Item Image labels now created for all photos automatically. Wide photos always placed last and validated correctly for column-3 skip logic. "Foundation plan" slides skipped automatically. Program 3: Item Image labels now created for any photo missing one. Text-only continuation slide created when body text overflows. All programs: photo size filter raised to ≥200px, tighter x/y bounds on stray text box cleanup.
v1.5	Resolved wide photo and text box placement bug. Wide photos now correctly consume 2 grid slots.
v1.4	Initial release to BEAR team. Program 2 updated to include arrow placement. All page/slide dialogs removed. Off-canvas image exclusion fixed. Page range auto-detection added to Program 4.
v1.3	Added rebar rusting slide detection. Reworked Condense Slides to top-to-bottom traversal. Added auto-detection of start/end pages. Hidden slides excluded from detection.
v1.2	Added open file detection across all programs. Fixed dialog focus. Keynote brought back to front after dialogs.
v1.1	Added Program 4 (Condense Slides).
v1.0	Initial release with Programs 1, 2, and 3. Portrait and landscape grid layout, text box cleanup, and arrow placement.